

FC7xxx CRC Integration Manual

Rev.0.6

Revision History

Revision	Date	Changes
0.1	2023/12/15	Initial release
0.6	2023/03/29	Updated for MCAL V0.6.0 - Added support for FC7240

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Chapter 1 Introduction

1.1 Introduction

This integration manual describes the integration requirements for CRC module.

Chapter 2 Building

2.1 Dependencies on Other Modules

Module configuration dependency

- Mcu: This module provides the clock reference point for CRC module.
- Common: This module is the basic module which used to choose the chip.

Module build dependency

- Common: This module provides common type for other modules.
- Rte: This module provides APIs to protect/unprotect some parts of code from interrupts (Exclusive Areas).
- Det: This module is necessary for enabling Development error detection.

Module initialization dependency

- Mcu: To use the CRC module, the user needs to initialize MCU module first.

2.2 Files Required for Compile

CRC module files:

- _MCAL/Src/Crc/src/Crc.c
- _MCAL/Src/Crc/src/Crc_Hw.c
- _MCAL/Src/Crc/src/Crc_Lookup_Tables.c
- _MCAL/Src/Crc/src/Crc_Reverse.c
- _MCAL/Src/Crc/src/Crc_RunTime.c
- _MCAL/Src/Crc/include/Crc.h
- _MCAL/Src/Crc/include/Crc_Hw.h
- _MCAL/Src/Crc/include/Crc_Lookup_Tables.h
- _MCAL/Src/Crc/include/Crc_MemMap.h
- _MCAL/Src/Crc/include/Crc_Reg.h
- _MCAL/Src/Crc/include/Crc_RegOps.h
- _MCAL/Src/Crc/include/Crc_Reverse.h
- _MCAL/Src/Crc/include/Crc_RunTime.h
- _MCAL/Src/Crc/include/Crc_Types.h
- _MCAL/Src/Crc/include/Crc_Version.h

- _MCAL_generate/src/Crc_Cfg.c
- _MCAL_generate/include/Crc_Cfg.h

Common module files:

- _MCAL/Src/Common/include/Mcal.h
- _MCAL/Src/Common/include/Std_Types.h
- MCAL/Src/Common/include/Platform_Types.h
- _MCAL/Src/Common/include/Compiler.h

- _MCAL/Src/Common/include/Compiler_Cfg.h
- _MCAL/Src/Common/include/CompilerDefinition.h

Det module files:

- Det.h

Ret module files:

- SchM_Crc.h

2.3 Add Plug-ins

CRC module plug-ins are developed for EB tresos Studio, so, to use CRC plug-ins on EB tresos Studio, the user needs to add the CRC module to the EB plug-ins folder first.

- 1) Copy CRC module (_MCAL/EB_Plugins/eclipse/plugins/CRC) folder to EB tresos plugins (EB/tresos/plugins/) folder.
- 2) Set CRC module output location folder for the generated source file and header files.
- 3) Use EB tresos to configure CRC module and generate source and header files.

Chapter 3 Memory

3.1 Sections in Memory Map

Section Name	Section Type	Description
CRC_START_SEC_CONFIG_DATA_8 CRC_STOP_SEC_CONFIG_DATA_8 CRC_START_SEC_CONFIG_DATA_16 CRC_STOP_SEC_CONFIG_DATA_16 CRC_START_SEC_CONFIG_DATA_32 CRC_STOP_SEC_CONFIG_DATA_32	Variables	This all sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code (data).
CRC_START_SEC_CONFIG_DATA_UNSPECIFIED CRC_STOP_SEC_CONFIG_DATA_UNSPECIFIED	Configuration Data	Start and stop of Memory Section for Config Data.
CRC_START_SEC_CONST_BOOLEAN CRC_STOP_SEC_CONST_BOOLEAN CRC_START_SEC_CONST_8 CRC_STOP_SEC_CONST_8 CRC_START_SEC_CONST_16 CRC_STOP_SEC_CONST_16 CRC_START_SEC_CONST_32 CRC_STOP_SEC_CONST_32 CRC_START_SEC_CONST_64 CRC_STOP_SEC_CONST_64 CRC_START_SEC_CONST_UNSPECIFIED CRC_STOP_SEC_CONST_UNSPECIFIED	Variables	This all sections used for variables which have to be aligned to 8/16/32 bit or boolean. These variables are read only (rodata).
CRC_START_SEC_CODE CRC_STOP_SEC_CODE CRC_START_SEC_RAMCODE CRC_STOP_SEC_RAMCODE CRC_START_SEC_CODE_AC CRC_STOP_SEC_CODE_AC	Code	Start and stop of memory Section for Code (text).
CRC_START_SEC_VAR_NO_INIT_BOOLEAN CRC_STOP_SEC_VAR_NO_INIT_BOOLEAN CRC_START_SEC_VAR_NO_INIT_8 CRC_STOP_SEC_VAR_NO_INIT_8 CRC_START_SEC_VAR_NO_INIT_16 CRC_STOP_SEC_VAR_NO_INIT_16 CRC_START_SEC_VAR_NO_INIT_32 CRC_STOP_SEC_VAR_NO_INIT_32 CRC_START_SEC_VAR_NO_INIT_UNSPECIFIED CRC_STOP_SEC_VAR_NO_INIT_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are never cleared and never initialized by startup code (bss).

Section Name	Section Type	Description
CRC_START_SEC_VAR_INIT_BOOLEAN CRC_STOP_SEC_VAR_INIT_BOOLEAN CRC_START_SEC_VAR_INIT_8 CRC_STOP_SEC_VAR_INIT_8 CRC_START_SEC_VAR_INIT_16 CRC_STOP_SEC_VAR_INIT_16 CRC_START_SEC_VAR_INIT_32 CRC_STOP_SEC_VAR_INIT_32 CRC_START_SEC_VAR_INIT_UNSPECIFIED CRC_STOP_SEC_VAR_INIT_UNSPECIFIED	Variables	These are all the sections used for variables which have to be aligned to 8/16/32 bit. These variables are initialized by startup code (data).

Chapter 4 Exclusive Area

None.

Chapter 5 Interrupt Service Routine (ISR)

None.

Chapter 6 Error Report

6.1 Det

Function Name	Error Type
Crc_CalculateChannel	CRC_E_ALREADY_INITIALIZED; CRC_E_INIT_FAILED
Crc_CalculateCRC8	CRC_E_INVALID_POINTER;
Crc_CalculateCRC8H2F	CRC_E_INVALID_POINTER;
Crc_CalculateCRC16	CRC_E_INVALID_POINTER;
Crc_CalculateCRC16ARC	CRC_E_INVALID_POINTER;
Crc_CalculateCRC32	CRC_E_INVALID_POINTER;
Crc_CalculateCRC32P4	CRC_E_INVALID_POINTER;
Crc_CalculateCRC64	CRC_E_INVALID_POINTER;
Crc_GetVersionInfo	CRC_E_INVALID_POINTER;

6.2 Dem

None

Chapter 7 Function Calls

7.1 Function Calls during Startup

7.2 Function Calls during Shutdown

None

7.3 Function Calls during Wake-up

None

7.4 Function Calls during Runtime

None

Chapter 8 Other Requirements

8.1 Notification, Callback, Callout

- Notification
None.
- Callback
None.
- Callout
None.

8.2 Macros

Please check various definitions available in Common module include file Mcal.h for details.

If OS not used:

If OS used:

Do not define AUTOSAR_OS_NOT_USED

Chapter 9 Integration Steps

- 1) Configure the CRC module and generate configuration files (Please refer to Building chapter for details).
- 2) Configure appropriate memory sections in linker file or other (Please refer to Memory chapter for details).
- 3) Build the CRC module with dependent modules.